

Lab Assignment # 07

Course Title : **AI Assistant Coding**
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Batch No. : **48**

Lab 7: Error Debugging with AI: Systematic approaches to finding and fixing bugs

Task Description #1 (Syntax Errors – Missing Parentheses in Print Statement)

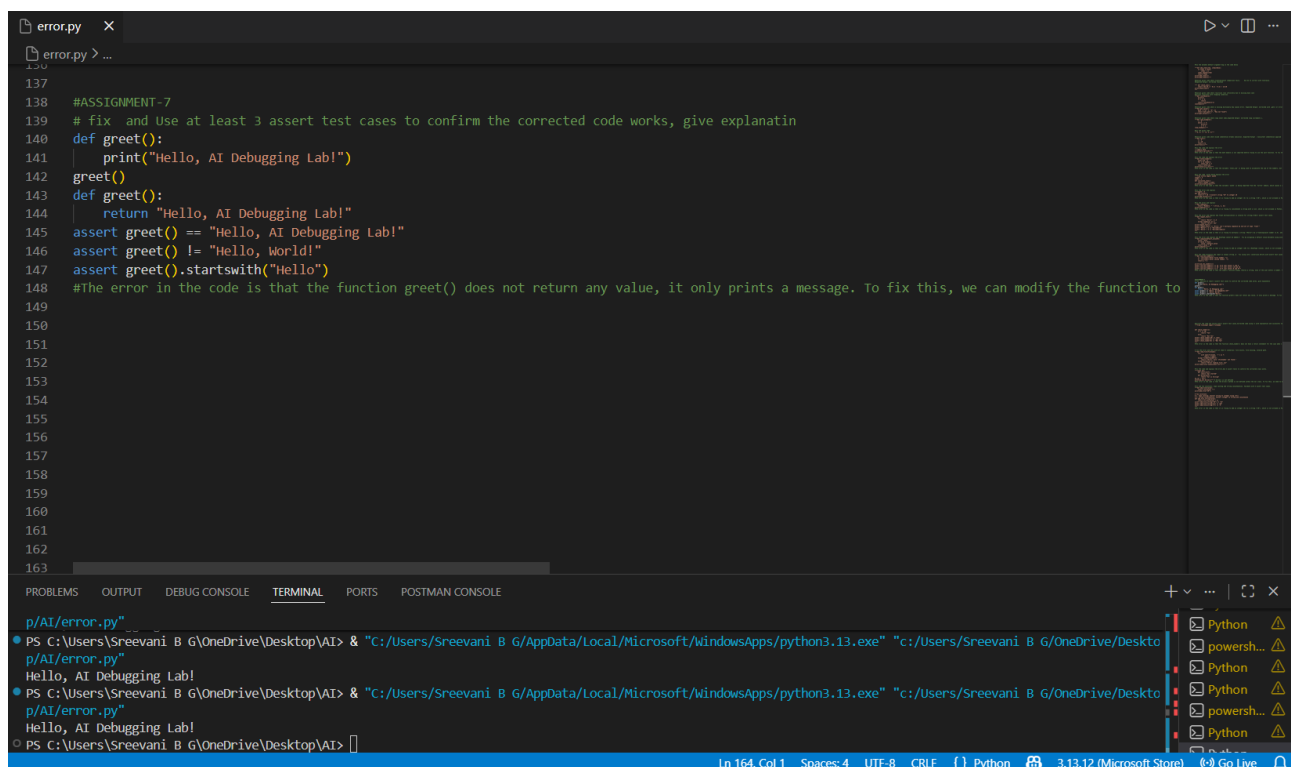
Task: Provide a Python snippet with a missing parenthesis in a print statement (e.g., print "Hello"). Use AI to detect and fix the syntax error.

Prompt: “Fix the missing parentheses in the Python print statement, then run the corrected code with at least 3 assert test cases to confirm it works..”

Expected Output #1:

- Corrected code with proper syntax and AI explanation.

Output Screenshot:



```
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138 #ASSIGNMENT-7
139 # fix and Use at least 3 assert test cases to confirm the corrected code works, give explanatin
140 def greet():
141     print("Hello, AI Debugging Lab!")
142 greet()
143 def greet():
144     return "Hello, AI Debugging Lab!"
145 assert greet() == "Hello, AI Debugging Lab!"
146 assert greet() != "Hello, World!"
147 assert greet().startswith("Hello")
148 #The error in the code is that the function greet() does not return any value, it only prints a message. To fix this, we can modify the function to
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PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS POSTMAN CONSOLE

p/Al/error.py"

PS C:\Users\Sreevani B G\OneDrive\Desktop\AI> & "C:/Users/Sreevani B G/AppData/Local/Microsoft/WindowsApps/python3.13.exe" "c:/Users/Sreevani B G/OneDrive/Deskto

p/Al/error.py"

Hello, AI Debugging Lab!

PS C:\Users\Sreevani B G\OneDrive\Desktop\AI> & "C:/Users/Sreevani B G/AppData/Local/Microsoft/WindowsApps/python3.13.exe" "c:/Users/Sreevani B G/OneDrive/Deskto

p/Al/error.py"

Hello, AI Debugging Lab!

PS C:\Users\Sreevani B G\OneDrive\Desktop\AI> |

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Task Description #2 (Incorrect condition in an If Statement)

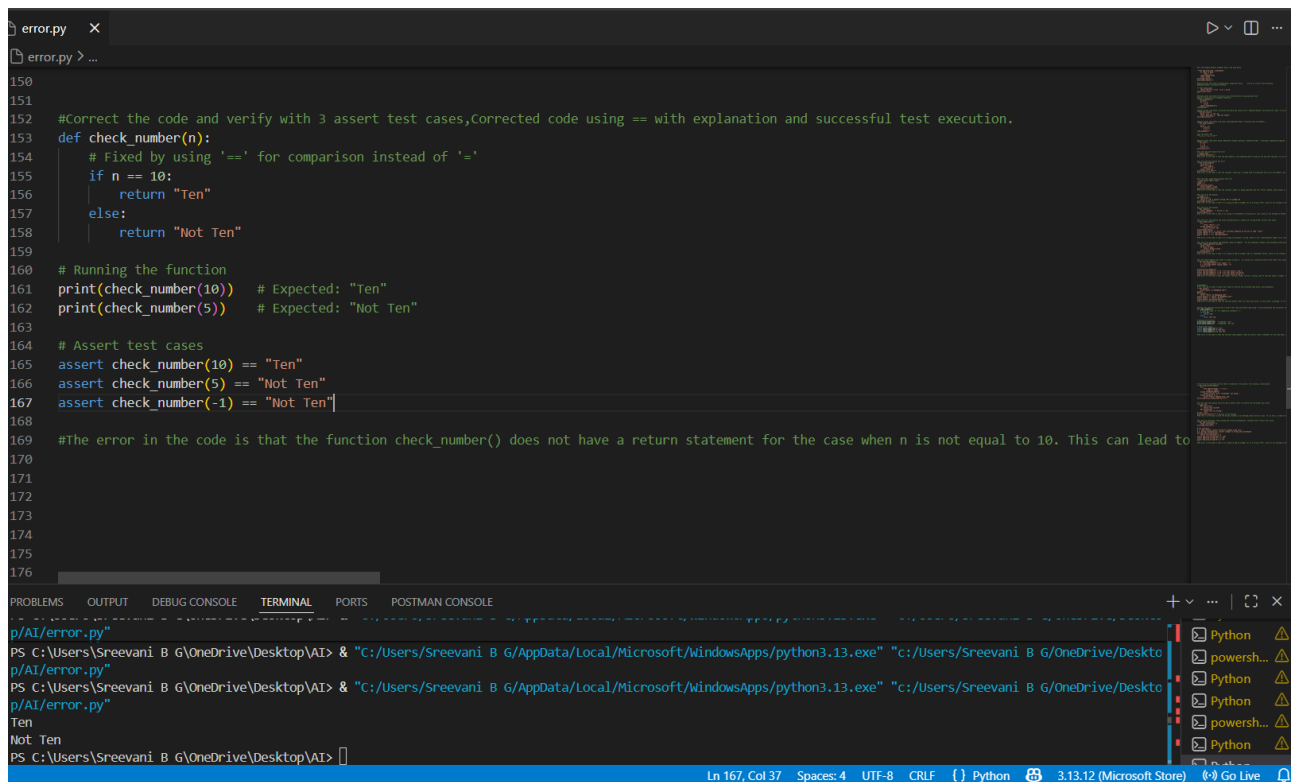
Task: Supply a function where an if-condition mistakenly uses = instead of ==. Let AI identify and fix the issue.

prompt: Fix the if-condition that mistakenly uses = instead of ==, then run the corrected code with at least 3 assert test cases to confirm it works.

Expected Output #2:

•Corrected code using == with explanation and successful test execution

Output Screenshot:



```
error.py
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152 #Correct the code and verify with 3 assert test cases,Corrected code using == with explanation and successful test execution.
153 def check_number(n):
154     # Fixed by using '=' for comparison instead of '='
155     if n == 10:
156         return "Ten"
157     else:
158         return "Not Ten"
159
160 # Running the function
161 print(check_number(10)) # Expected: "Ten"
162 print(check_number(5)) # Expected: "Not Ten"
163
164 # Assert test cases
165 assert check_number(10) == "Ten"
166 assert check_number(5) == "Not Ten"
167 assert check_number(-1) == "Not Ten"
168
169 #The error in the code is that the function check_number() does not have a return statement for the case when n is not equal to 10. This can lead to
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```

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PS C:\Users\Sreevani B G\OneDrive\Desktop\AI> & "C:/Users/Sreevani B G/AppData/Local/Microsoft/windowsApps/python3.13.exe" "c:/Users/Sreevani B G/OneDrive/Desktop/p/AI/error.py"
Ten
Not Ten
PS C:\Users\Sreevani B G\OneDrive\Desktop\AI>
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Ln 167, Col 37 Spaces: 4 UTF-8 CRLF Python 3.13.12 (Microsoft Store) Go Live

Task Description #3 (Runtime Error – File Not Found)

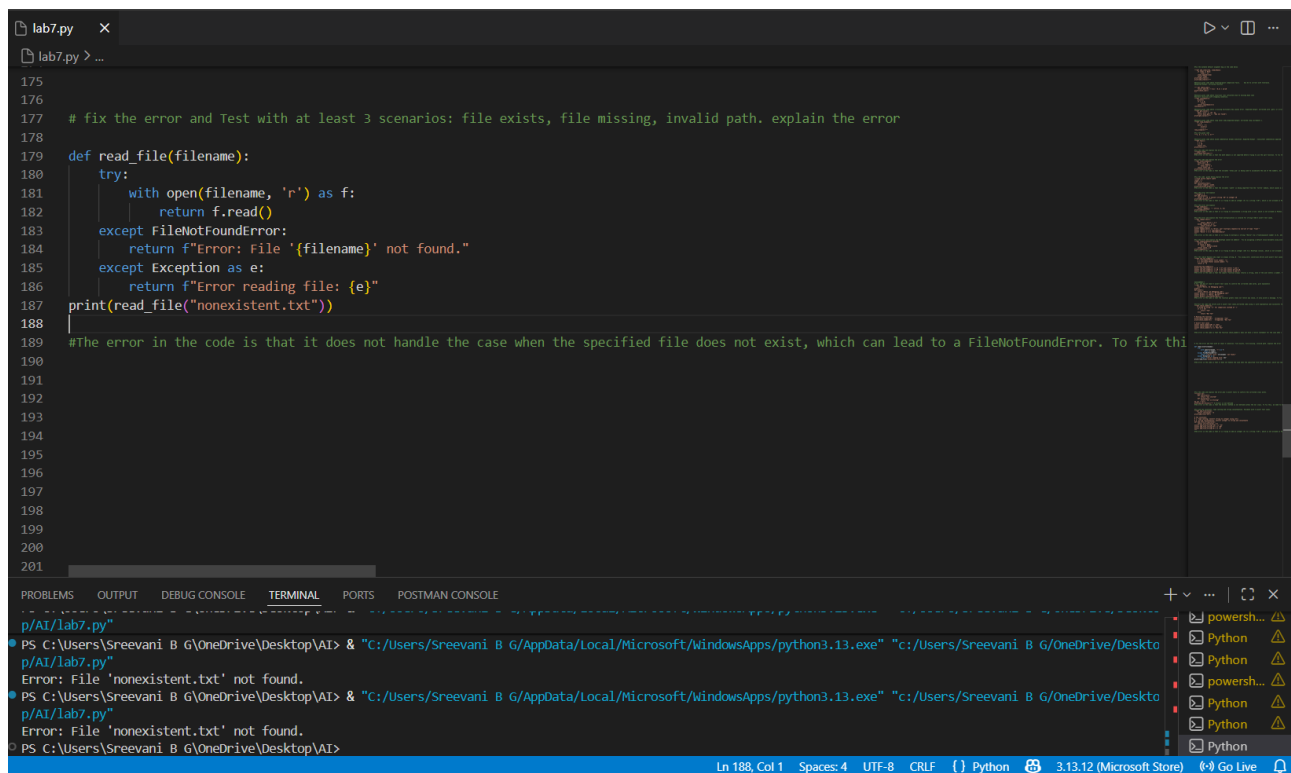
Task: Provide code that attempts to open a non-existent file and crashes. Use AI to apply safe error handling.

Prompt: "Use try-except to handle missing file errors with a message and test with 3 scenarios."

Expected Output #3:

- Safe file handling with exception management.

Output Screenshot:



The screenshot shows a VS Code editor window with a file named `lab7.py`. The code defines a function `read_file(filename)` that attempts to open a file in read mode. It uses a `try-except` block to handle `FileNotFoundError` and other exceptions. The function returns an error message if the file is not found or if an exception occurs while reading. The code is tested with the filename `nonexistent.txt`.

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177 # fix the error and Test with at least 3 scenarios: file exists, file missing, invalid path. explain the error
178
179 def read_file(filename):
180     try:
181         with open(filename, 'r') as f:
182             return f.read()
183     except FileNotFoundError:
184         return f"Error: File '{filename}' not found."
185     except Exception as e:
186         return f"Error reading file: {e}"
187 print(read_file("nonexistent.txt"))
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189 #The error in the code is that it does not handle the case when the specified file does not exist, which can lead to a FileNotFoundError. To fix thi
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The terminal output shows the execution of the script, which results in the error message: `Error: File 'nonexistent.txt' not found.`

```
p/Al/lab7.py"
PS C:\Users\Sreevani B G\OneDrive\Desktop\AI> & "C:/Users/Sreevani B G/AppData/Local/Microsoft/WindowsApps/python3.13.exe" "c:/Users/Sreevani B G/OneDrive/Deskto
p/Al/lab7.py"
Error: File 'nonexistent.txt' not found.
PS C:\Users\Sreevani B G\OneDrive\Desktop\AI> & "C:/Users/Sreevani B G/AppData/Local/Microsoft/WindowsApps/python3.13.exe" "c:/Users/Sreevani B G/OneDrive/Deskto
p/Al/lab7.py"
Error: File 'nonexistent.txt' not found.
PS C:\Users\Sreevani B G\OneDrive\Desktop\AI>
```

The status bar at the bottom indicates the current line and column (Ln 188, Col 1), the number of spaces (4), the encoding (UTF-8), the line ending (CRLF), the language (Python), and the version (3.13.12 (Microsoft Store)).

Task Description #4 (Calling a Non-Existent Method)

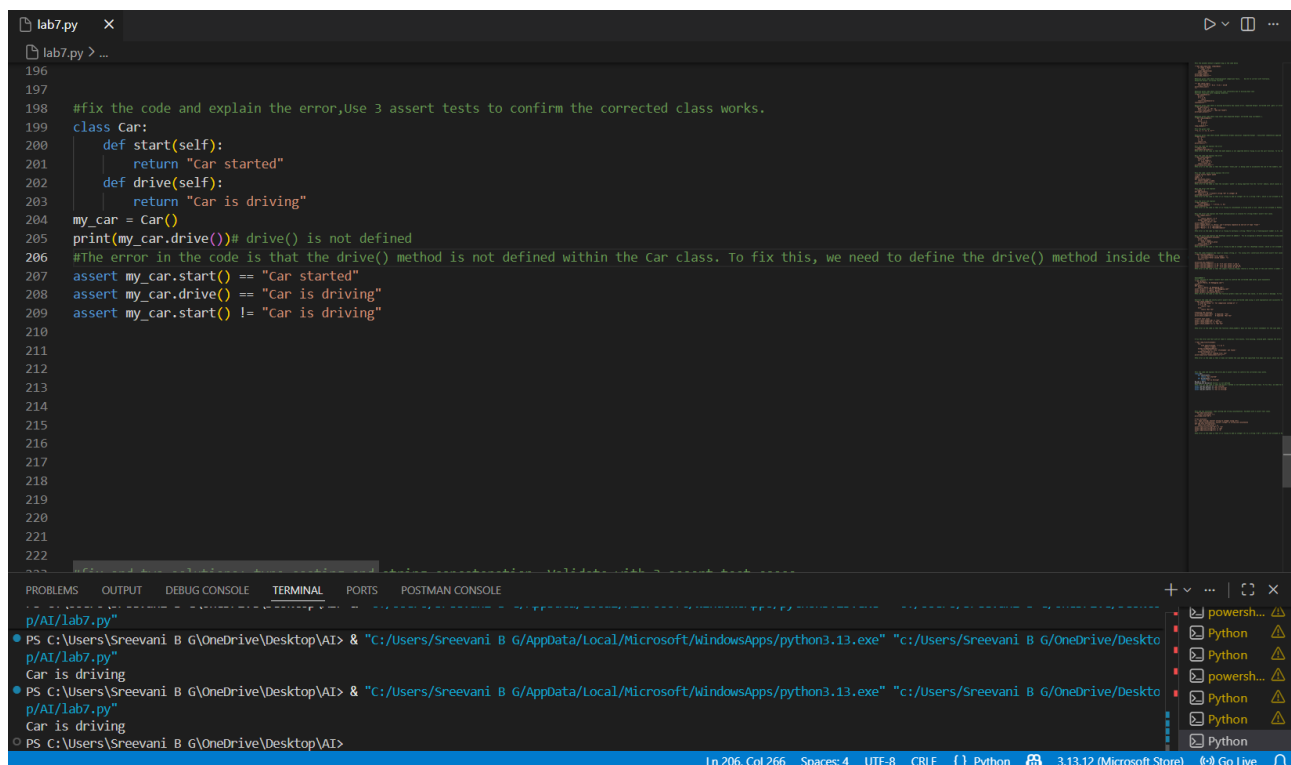
Task: Give a class where a non-existent method is called (e.g., `obj.undefined_method()`). Use AI to debug and fix.

Prompt: “Fix the Car class bug where `drive()` is called but not defined, and validate the corrected class with 3 assert tests.”

Expected Output #4:

- Corrected class with clear AI explanation.

Output Screenshot:



```
lab7.py
lab7.py > ...
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197
198 #fix the code and explain the error,Use 3 assert tests to confirm the corrected class works.
199 class Car:
200     def start(self):
201         return "Car started"
202     def drive(self):
203         return "Car is driving"
204 my_car = Car()
205 print(my_car.drive())# drive() is not defined
206 #The error in the code is that the drive() method is not defined within the Car class. To fix this, we need to define the drive() method inside the
207 assert my_car.start() == "Car started"
208 assert my_car.drive() == "Car is driving"
209 assert my_car.start() != "Car is driving"
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PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS POSTMAN CONSOLE
p/Al/lab7.py
PS C:\Users\Sreevani B G\OneDrive\Desktop\AI> & "C:/Users/Sreevani B G/AppData/Local/Microsoft/WindowsApps/python3.13.exe" "C:/Users/Sreevani B G/OneDrive/Desktop/Al/lab7.py"
Car is driving
PS C:\Users\Sreevani B G\OneDrive\Desktop\AI> & "C:/Users/Sreevani B G/AppData/Local/Microsoft/WindowsApps/python3.13.exe" "C:/Users/Sreevani B G/OneDrive/Desktop/Al/lab7.py"
Car is driving
PS C:\Users\Sreevani B G\OneDrive\Desktop\AI>
```

Ln 206, Col 266 Spaces: 4 UTF-8 CRLF Python 3.13.12 (Microsoft Store) Go Live

Task Description #5 (TypeError – Mixing Strings and Integers in Addition)

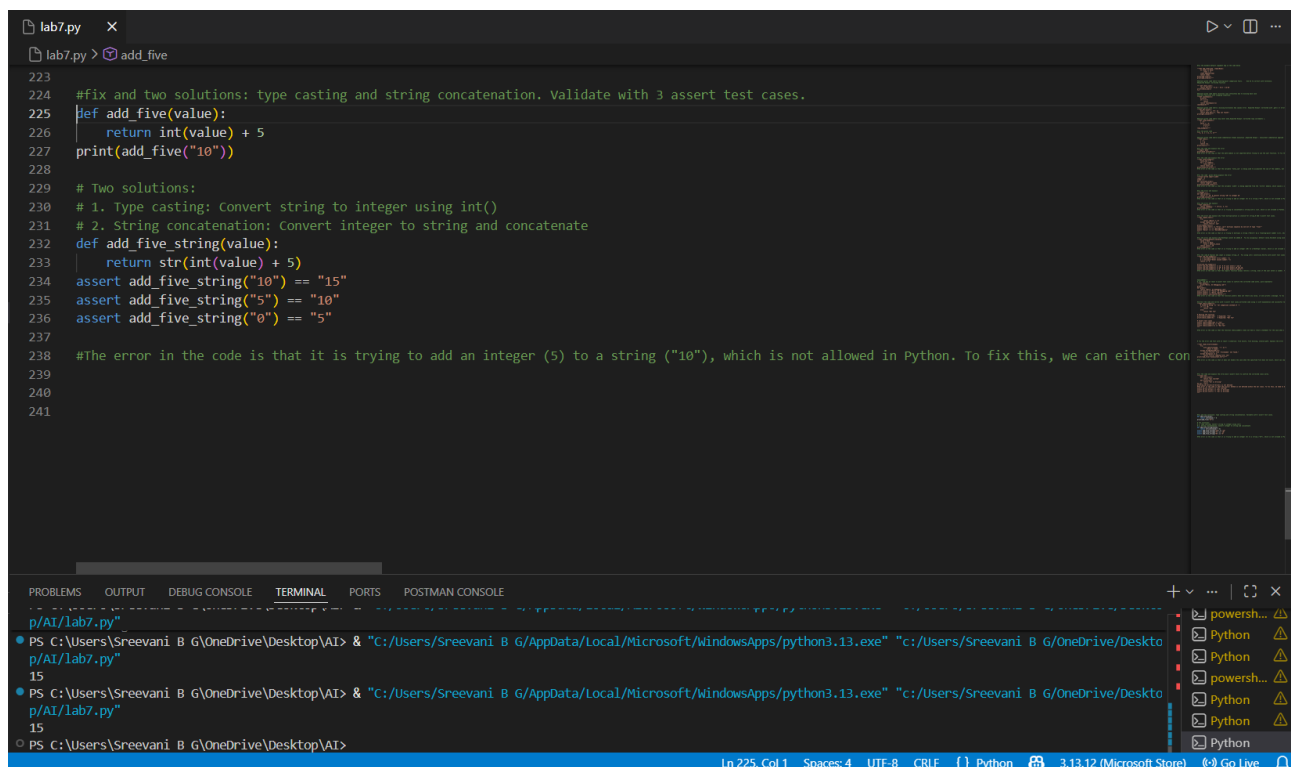
Task: Provide code that adds an integer and string ("5" + 2) causing a TypeError. Use AI to resolve the bug.

Prompt: “Fix the TypeError in add_five("10") by providing two solutions (type casting and string concatenation) and validate with 3 assert tests.”

Expected Output #5:

- Corrected code that runs successfully for multiple inputs.

Output Screenshot:



```
lab7.py
lab7.py > add_five

223
224 #fix and two solutions: type casting and string concatenation. Validate with 3 assert test cases.
225 def add_five(value):
226     return int(value) + 5
227 print(add_five("10"))
228
229 # Two solutions:
230 # 1. Type casting: Convert string to integer using int()
231 # 2. String concatenation: Convert integer to string and concatenate
232 def add_five_string(value):
233     return str(int(value) + 5)
234 assert add_five_string("10") == "15"
235 assert add_five_string("5") == "10"
236 assert add_five_string("0") == "5"
237
238 #The error in the code is that it is trying to add an integer (5) to a string ("10"), which is not allowed in Python. To fix this, we can either con
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PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS POSTMAN CONSOLE

```
PS C:\Users\Sreevani B G\OneDrive\Desktop\AI> & "C:/Users/Sreevani B G/AppData/Local/Microsoft/WindowsApps/python3.13.exe" "c:/Users/Sreevani B G/OneDrive/Deskto
p/AI/lab7.py"
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PS C:\Users\Sreevani B G\OneDrive\Desktop\AI> & "C:/Users/Sreevani B G/AppData/Local/Microsoft/WindowsApps/python3.13.exe" "c:/Users/Sreevani B G/OneDrive/Deskto
p/AI/lab7.py"
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PS C:\Users\Sreevani B G\OneDrive\Desktop\AI>
```

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